



GANESH REDDY

Gerhart-Hauptman-str. 11, Erlangen, 91058

+49 1551 0182395

| ganeshreddy30102000@gmail.com |

[Portfolio](#) | [LinkedIn](#) | [GitHub](#)

PROFESSIONAL SUMMARY

Master's student in **Autonomy Technologies** with hands-on experience in **robotics software (ROS2, Python/C++)**, **simulation (Gazebo)**, **computer vision (OpenCV)**, and **engineering data pipelines**. Strong foundation in building and validating technical systems through **structured testing, debugging, and documentation**. Interested in practical development at the intersection of **vehicle control, monitoring, and system integration** for cooperative driving demonstrators.

EDUCATION

FRIEDRICH-ALEXANDER UNIVERSITY

Erlangen, Germany

Master of Autonomy Technology

Apr 2024 – Present

- **Relevant coursework:** Introduction to Deep Learning, Pattern Recognition, Computer Vision, Machine Learning in Signal Processing, AI in Medical Robotics, Robotics in Surgery and Diagnostics, Human-Robot Co-adaptation.
- **GPA:** 2.7 (ECTS: 62.5/120)

KLE TECHNOLOGICAL UNIVERSITY

Hubballi, India

Bachelor of Automation and Robotics

Aug 2018 – Jul 2022

- **Focus areas:** Machine Learning, Signal Processing, Data structures.
- **GPA:** 2.3
- **Bachelor's Thesis:** Automated Grading & Sorting System for Vegetables and Fruits.

PROFESSIONAL EXPERIENCE

Friedrich-Alexander-Universität Erlangen–Nürnberg (FAU)

Erlangen, Germany

Student Research Assistant- Part time

Nov 2025 – present

- Create Python-based exercises and automated evaluation scripts for teaching/research datasets, helping reduce manual checking effort by **~30%** and improving consistency of feedback.
- Solve and clarify students' doubts in Python/data workflow tasks, reducing repeated clarification requests by **~25%** through structured examples and debugging support.
- Support data quality validation and structured reporting for assignments/experiments, improving turnaround time for evaluation/report generation by **~20%**.

Digital Associates India

Hyderabad, India

Internet of Things (IoT) Intern

Jan 2023 – Oct 2023

- Conducted functional testing of ESP32-based devices and validated components using oscilloscopes/multimeters, helping reduce hardware debugging rework by **~20%** through structured test checks.
- Designed custom PCBs in KiCad and optimized sensor integration for reliable data acquisition in embedded systems.
- Developed BLE-based indoor localization (RSSI-based) for medical asset tracking use cases and documented test setups/milestones in Agile workflows (Jira, Gantt tracking).

Centre for Autonomous Systems Engineering

Hubballi, India

Systems Engineer Intern

Jan 2022 – Jul 2022

- Designed and fabricated a **5-DOF robotic arm** (AGV context) in SolidWorks and produced manufacturing drawings for prototype development.
- Built URDF-based robot models and validated kinematics in **ROS/ROS2 + Gazebo**, integrated **MoveIt** for motion planning and trajectory execution.
- Supported simulation-first verification and technical documentation, improving development handoff clarity and reducing iteration delays in team workflows.

TECHNICAL COMPETENCIES

- **Programming:** Python, C++, C, MATLAB.
- **Control and Modelling:** Kinematic modeling, motion analysis, trajectory execution, PID control, parameter tuning.
- **Robotics Simulation & Motion Planning:** MuJoCo, ROS/ROS2, Gazebo, MoveIt, URDF/Xacro, RViz, Nav2, AMCL.
- **Simulation & Analysis:** MATLAB/Simulink, simulation-based testing, structured debugging, reproducible validation workflows.
- **Computer Vision & AI:** OpenCV, PyTorch, TensorFlow, YOLOv3, scikit-learn.
- **Embedded Systems:** ESP32 (ESP-IDF), Arduino, Raspberry Pi, MQTT, sensor integration.
- **CAD & Prototyping:** SolidWorks, 3D printing, MATLAB/Simulink.
- **Tools:** Git, Linux (Ubuntu), PlatformIO, Power BI, Jira, Microsoft Office.

PROJECTS

5-DOF Robotic Arm Simulation & Motion Planning (ROS/ROS2, Gazebo, MoveIt, Mujoco)

- Built URDF-based robot model and validated kinematic behavior in Gazebo.
- Integrated MoveIt for motion planning and trajectory execution; performed motion analysis and iterative tuning.

SLAM & Navigation with TurtleBot (ROS2, Nav2, AMCL, Gazebo, RViz)

- Implemented ROS2-based SLAM and configured Nav2 + AMCL, tuned YAML parameters and validated localization in simulation.
- Tuned YAML parameters and validated localization/path behavior through repeatable simulation runs.
- Debugged navigation behavior in RViz/Gazebo and documented parameter impacts for reproducible testing.

ROS2 Turtlesim Bot Simulation (ROS2)

- Created custom ROS2 nodes for real-time control and master-slave coordination, implemented service calls for reset/teleport tasks.
- Implemented service-based control logic (spawn/reset/teleport) and reusable package structures.

Automated Grading & Sorting System

- Designed a conveyor-based mechanical sorting concept in SolidWorks and led a **5-member team** during implementation.
- Built a Python/OpenCV-based classification workflow for automation-oriented sorting, achieving 80% classification accuracy in prototype testing.
- Supported sensor-actuator coordination and documented failure modes to improve repeatability and robustness.

Quadcopter Simulation & Control (MATLAB/Simulink, SolidWorks, ROS/Gazebo)

- Simulated PID control loops in Simulink and validated URDF-based behavior in Gazebo for stability analysis.
- Used simulation for stability analysis and controller tuning

CERTIFICATIONS

- ROS for beginners I & II – SLAM, Navigation, Path Planning.
- Python from University of Michigan (Coursera)
- Full Stack Web development (Udemy).

LANGUAGES

- **English:** C1
- **German:** A2
- **Hindi:** C1

Availability: Immediate Internship or Working student or HiWi (up to 20h/week) for 12+ months (flexible), Available for **Master's thesis / project study (PA/MA)** and related lab/research work. Can start **as soon as possible**.

Ganesh Reddy